SOFTWARE INTRODUCTION

ShanweiGeo

3D Geoscience Data Management and Visualization

Project by Yingchun Rao | October 2026

ShanweiGeo is a local desktop workspace for organizing exploration data, viewing existing geological models and connecting three-dimensional scenes with maps, drillhole records and documents. It is designed around the deposit, exploration area and section used in geological work.

The current platform brings data organization, spatial inspection and presentation into one working environment. Its main contribution is a practical foundation for reviewing and communicating multi-source geoscience information.

Research Areas and the 3D Workspace

Data are organized in a three-level hierarchy: deposit, area and section. Each research region has its own data tree. Users can create, rename and remove regions, move data between them, and identify the region associated with the active view.

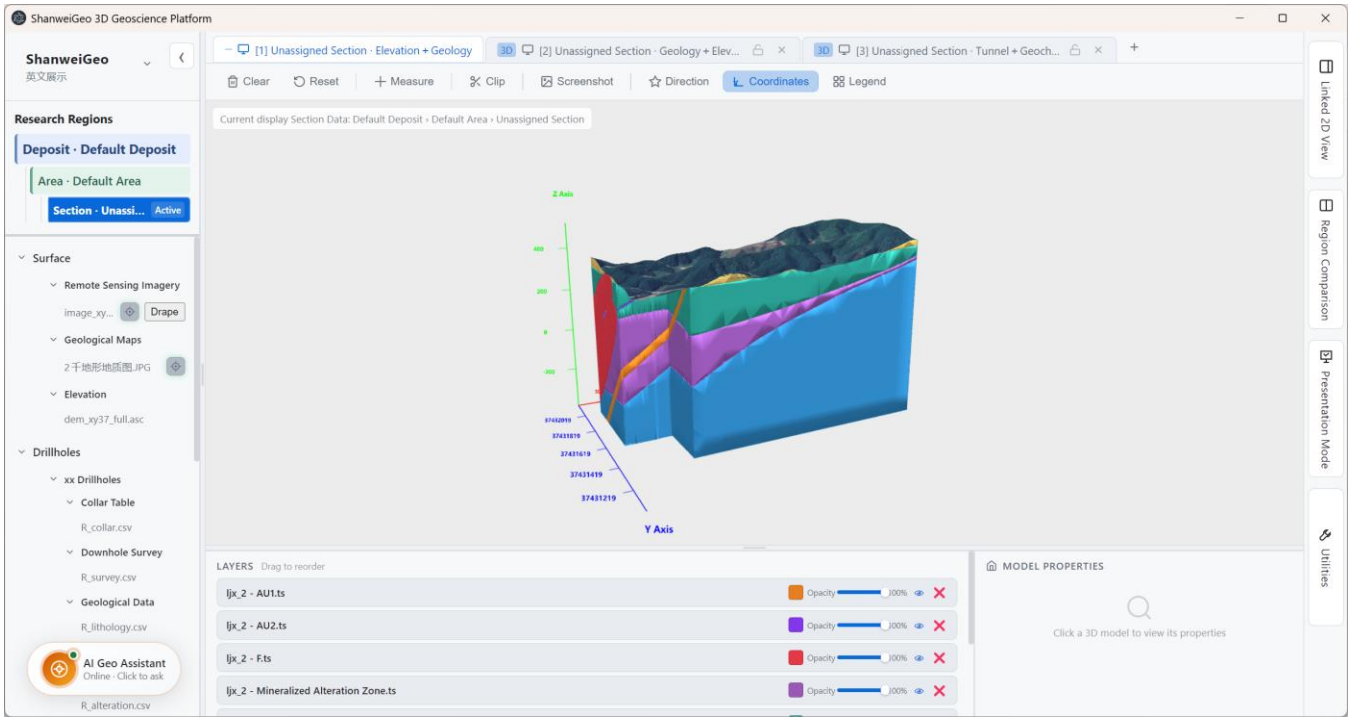


Figure 1. A geological model with terrain and draped imagery, alongside the research hierarchy, data tree and layer controls.

Keep different research contexts separate

Multiple 3D tabs maintain their own scenes, coordinate origins and cameras. Selecting a tab and selecting its corresponding item in the project tree support navigation in both directions. This allows several areas or model combinations to remain open during a review.

Control what the scene shows

The layer panel provides visibility, color, opacity and ordering controls. Users can select models, inspect their properties, adjust point or line size, and use rotation, panning, zooming and fit-to-view to examine spatial relationships.

Drillholes and Linked Evidence

Drillhole tables are managed alongside three-dimensional views. Lithological intervals can be displayed along drillhole trajectories, while the linked panel provides access to logs, core images and assay records for the selected hole.

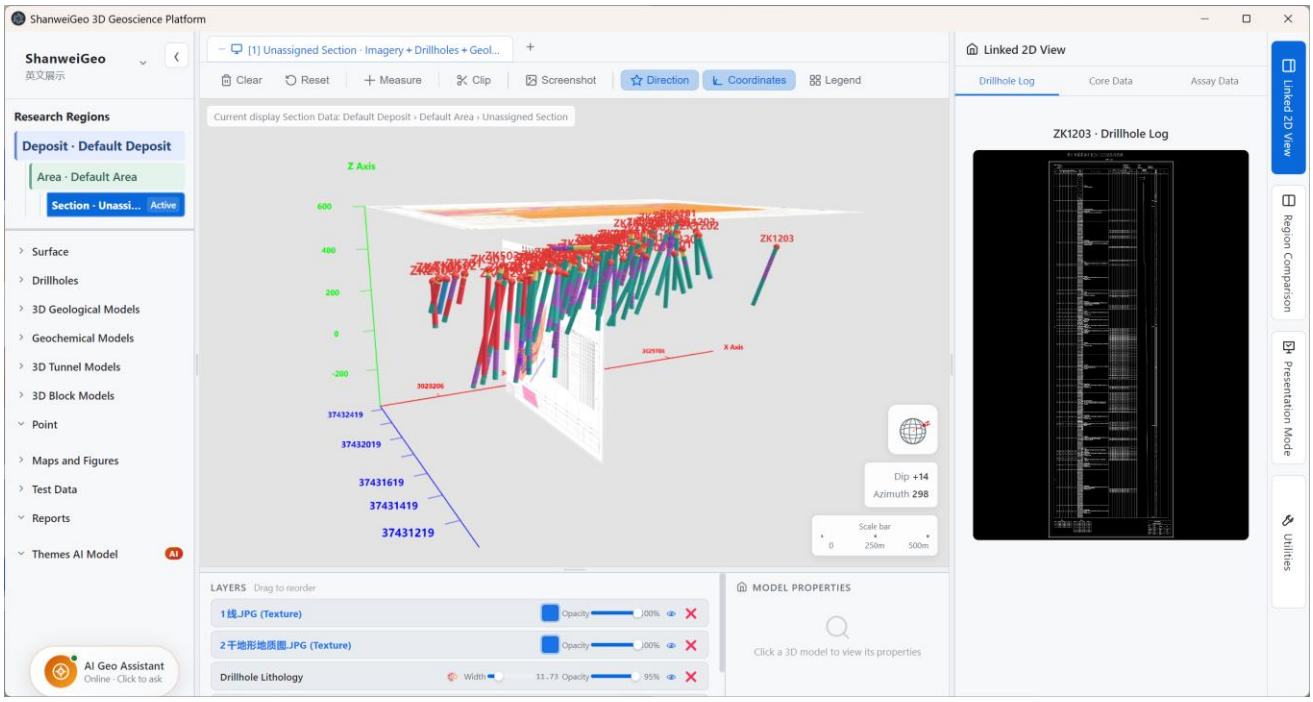


Figure 2. Drillholes and registered figures displayed together, with a drillhole log in the linked panel.



Figure 3. Core images in the linked panel at left and interval assay records at right.

This connection helps a reviewer move from the spatial position of a drillhole to its supporting observations. Depth intervals, field definitions and assay units should be checked against the original records when interpreting the data.

Geochemical Attributes in 3D

Geochemical attributes can be viewed along drillhole intervals using a numeric color scale. Users can choose the attribute to display and adjust the corresponding layer, making it easier to inspect the spatial distribution of measured values.

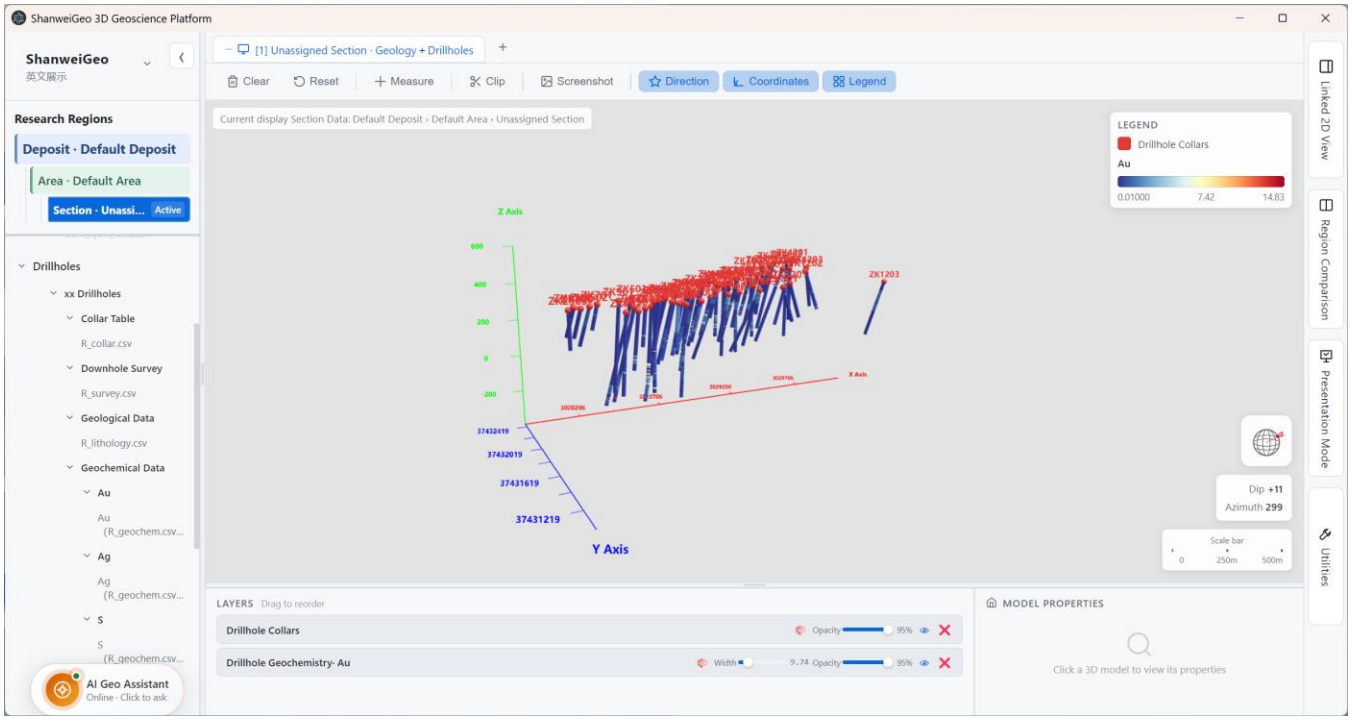


Figure 4. Drillholes colored by the Au attribute, with a continuous color legend and separate collar and geochemistry layers.

Read values in their geological context

A color scale makes relative variation visible in the scene. The original tables remain available for checking individual records, and the layer controls allow geochemical intervals to be viewed with geological models and other evidence.

Distinguish display from prediction

This view represents imported geochemical data. Displaying an attribute along drillholes does not by itself produce a new interpolated grade model or a prospectivity prediction. Those analyses require their own methods, assumptions and validation.

Maps and Spatial Registration

Maps and figures can be opened in their own tabs for inspection. Control-point registration relates image positions to model-space coordinates, allowing a registered figure to be displayed as an image plane in a 3D scene.

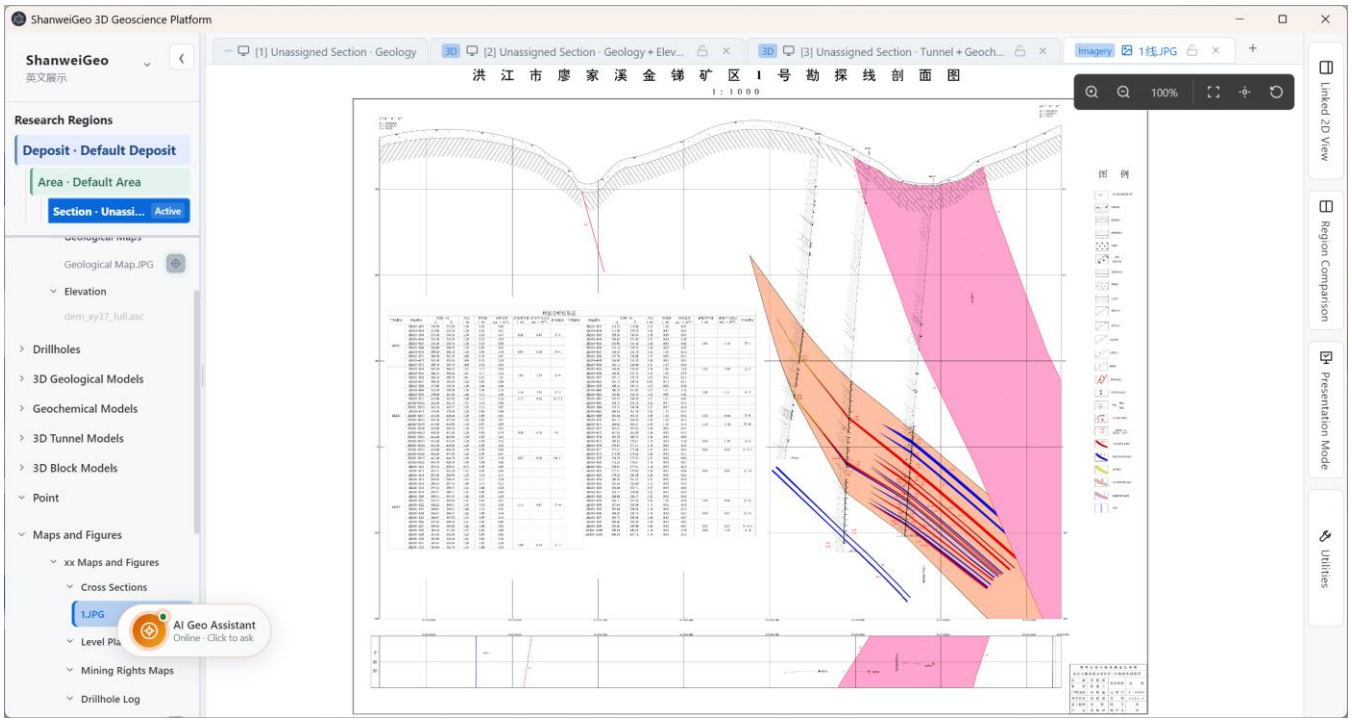


Figure 5. A geological cross-section open in the image viewer, with zoom, fit and reset controls.

Connect existing figures with the scene

The platform supports image and TIFF preview, control-point registration, and display of imagery with existing georeferencing. Georeferenced imagery can also be draped over elevation terrain. These tools help bring surface information and interpreted sections into the same environment as subsurface models.

Check spatial consistency

Before combining datasets, confirm their coordinate reference systems, units, horizontal extents and elevation references. A visually aligned scene is useful for inspection, but the registration result remains dependent on the source data and chosen control points.

Tables and Document Reading

Excel and CSV tables, PDF files and Word documents can be opened in dedicated tabs. Reading tools include PDF text selection, bookmarks and notes, so supporting records can be reviewed without leaving the workspace.

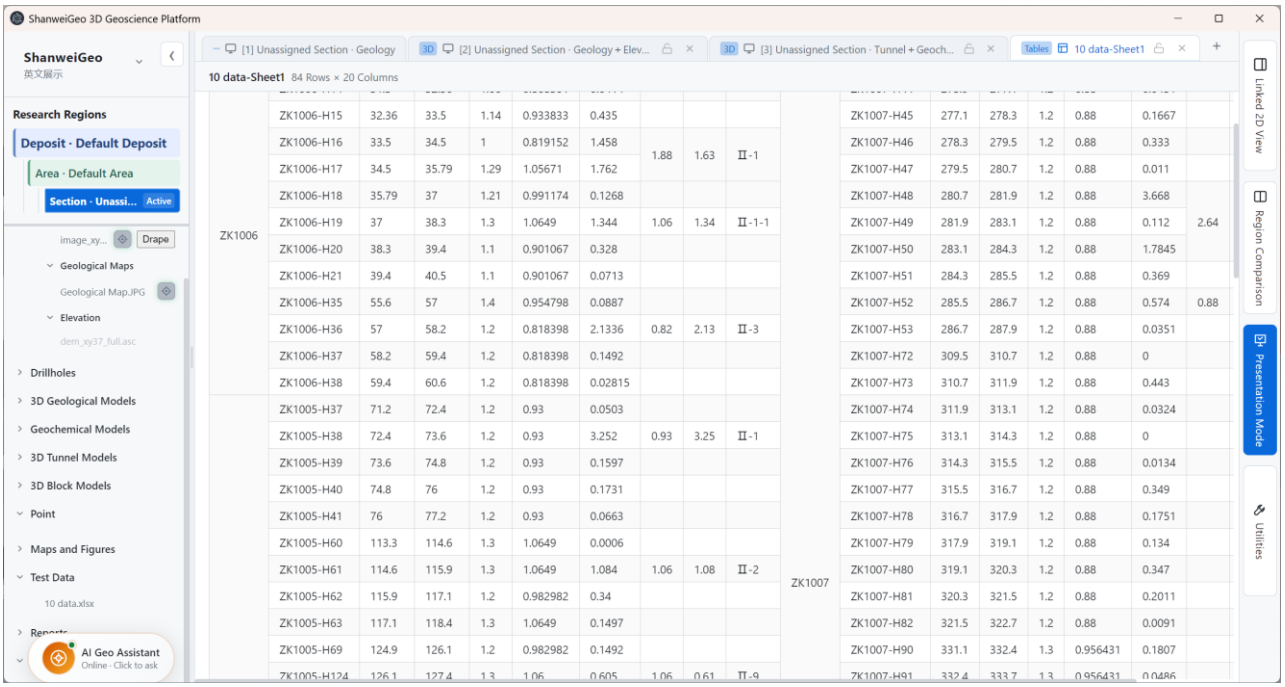


Figure 6. An exploration data table open in a tab within the same project workspace.

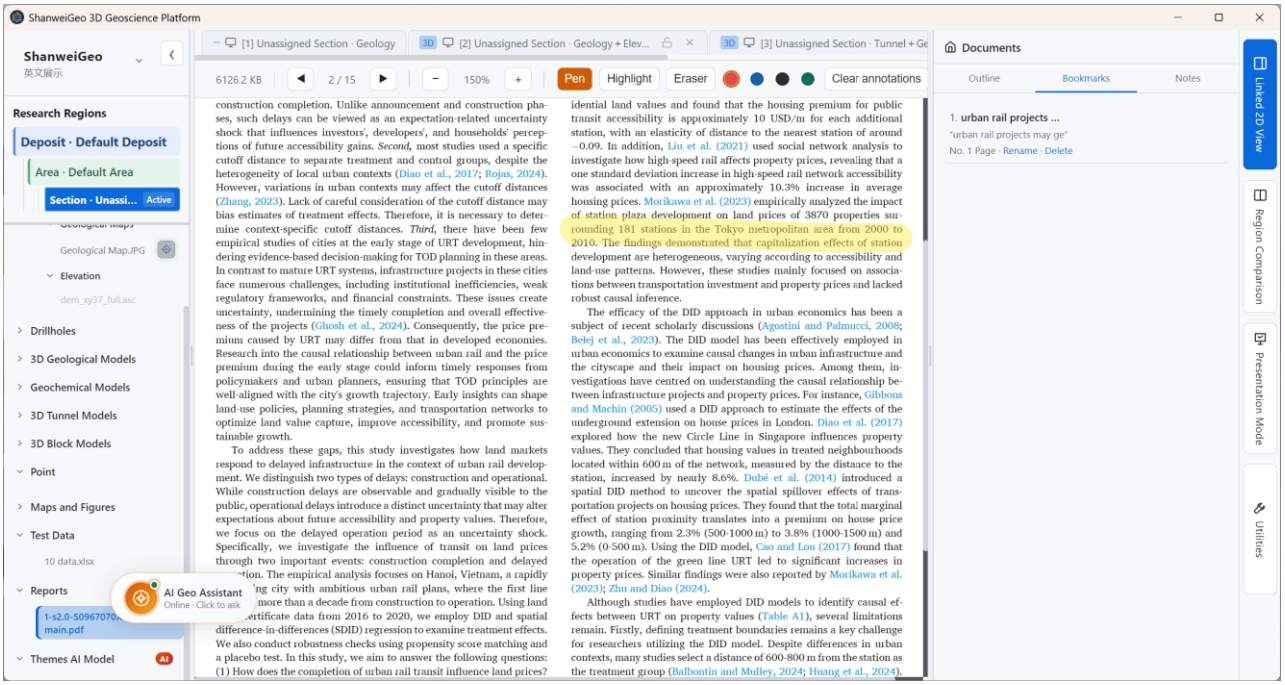


Figure 7. PDF reading with highlighting and a saved bookmark. The document is shown as a reading example.

Side by Side Model Comparison

Region Comparison displays two open 3D views next to each other. Each pane retains its own layer controls. Users can select the views, swap panes, adjust their relative width and fit both scenes to the available space.

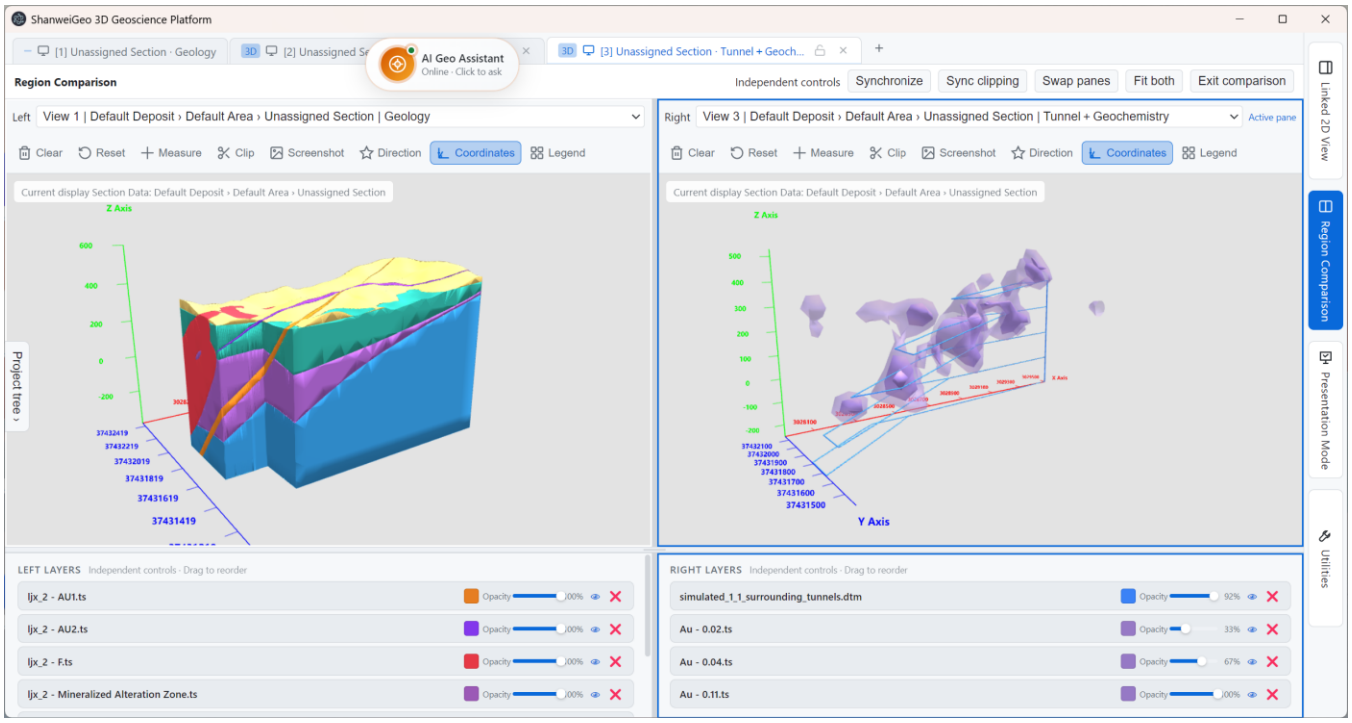


Figure 8. A geological model at left and tunnel and geochemical models at right in the comparison workspace.

Use independent or synchronized controls

Independent controls allow each scene to be inspected separately. Synchronized operation supports coordinated viewing, while synchronized clipping helps examine corresponding positions when the datasets share a suitable coordinate basis.

Discuss models and supporting observations

The comparison view can support geological discussion across model combinations, sections or research regions. It provides a visual comparison; it does not automatically align different coordinate systems or quantify differences between models.

Materials and Presentation

Presentation settings include soft lighting, matte materials and background controls. A local rock material library and custom textures can be applied to individual models, with settings available for adjusting their appearance.

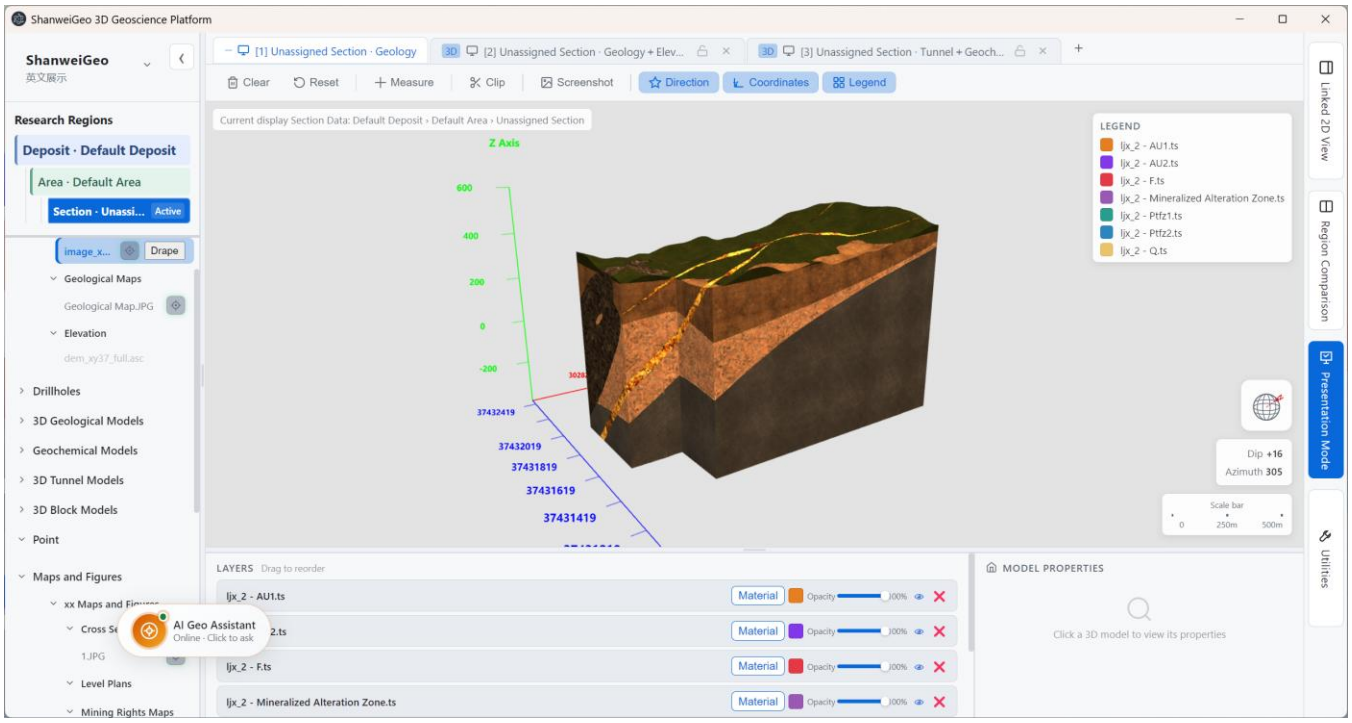


Figure 9. The model displayed with rock textures and terrain, with material controls available in the layer list.

Make the model easier to explain

Colors, opacity and textures can help distinguish model components and communicate their geometry. Coordinate axes, legends, screenshots and viewpoint controls support presentations and discussions of the three-dimensional scene.

Keep display choices separate from geological attributes

A texture is a visual representation. Applying it does not identify a rock type or verify the interpretation of a geological unit. Model attributes and their supporting observations remain the basis for geological interpretation.

Supporting Functions and Local Workflow

Additional implemented functions extend the workspace beyond the views illustrated above.

Module	Current capability
Model formats	TS and OBJ surfaces, paired STR/DTM tunnel models, and CSV block models.
Terrain and imagery	ASC elevation import, elevation colors, boundary-zero handling, vertical display scale, and imagery draping.
Points and lines	Point and line import, display controls, and integration with the research data tree.
Spatial tools	Measurement, clipping, screenshots, coordinate axes, legends and view reset.
Coordinate utilities	Coordinate conversion interface and parameter settings, with an entry point for the original COORD application.
Project persistence	Save and restore research structure, data ownership, tabs, layers, viewpoints and registration state; recover session drafts.

Module	Current capability
Local processing	Electron desktop interface and a local Python processing channel; document libraries loaded on demand and inactive views restored lazily.

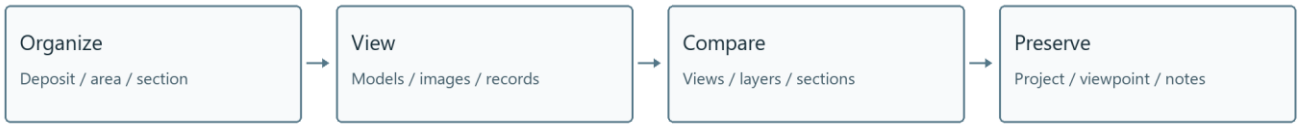


Figure 10. A typical workflow supported by the platform.

A foundation for further research

The current emphasis is data management, visualization of existing models and interactive inspection. This provides a starting point for future work on multi-source data integration and three-dimensional mineral exploration research. Automated geological modeling and mineral prospectivity prediction are separate development objectives.

Planned Research on Mineral Distribution

My next research step is to build on the deposit-specific 3D geological model I have established. I plan to integrate laboratory sample measurements with information derived from this model and use machine learning to quantitatively characterize the mineralogical and geochemical properties of the subsurface mineral system in three dimensions.

Integrate sample measurements and model information

I will link sample locations and their measured properties to the 3D geological framework, creating a common spatial basis for combining laboratory observations with geological model attributes. This integrated dataset will connect measured mineralogical and geochemical information with its geological setting.

Develop predictive models in three dimensions

Using the combined sample and model data, I plan to train machine-learning models to identify minerals and predict their occurrence and spatial distribution within the deposit system. The aim is to move from displaying existing geological information toward quantitative, deposit-scale characterization and prediction.

Connect predictions with geological evidence

A key methodological step will be to evaluate predictions against sample observations and their geological context. Predicted properties can then be returned to the 3D workspace for inspection alongside the original samples and models, allowing the results to be examined spatially.

Use the platform as a research foundation

ShanweiGeo provides the data management and visualization environment for this work. Sample-model integration, machine-learning training and three-dimensional mineral prediction form the planned next stage of the research; they are not presented here as completed software capabilities.